

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE AND EVIDENCE-BASED SYNTHESIS REPORT

1. ABSTRACT

This report analyzes patent evidence relevant to nitrile butadiene rubber (NBR) with emphasis on low-acrylonitrile embodiments, using the provided manifest patent US11347156B2 as the sole primary source. The patent is not a synthesis patent for NBR production; instead, it discloses an electro-conductive member architecture in which NBR is used as one of the diene-based rubbers in a matrix-domain rubber composite. The extracted evidence is still relevant to low-acrylonitrile NBR because it explicitly identifies several commercial NBR grades with nitrile contents spanning 15% to 41.5%, including a low-nitrile example at 15% and another at 18.0%, and it compares these materials in terms of solubility parameter, viscoelastic response, and electro-conductive member performance. The patent's core technical teaching is that pairing different diene rubbers, including NBR and SBR, can create a matrix-domain structure with improved mechanical recovery and stable discharge behavior when the SP-value difference and $\tan \delta$ ratio are controlled. The evidence also shows that NBR can be used either as the matrix or as the domain rubber, but the patent does not disclose NBR polymerization conditions, monomer feed ratios, initiator systems, or emulsion/solution synthesis routes. Accordingly, the report treats the patent as an indirect source for low-acrylonitrile NBR composition benchmarking rather than a direct synthesis disclosure. The major formulation trend is the use of NBR as a commercially supplied rubber grade with specified nitrile content, combined with carbon black, sulfur, and vulcanization accelerators to build electro-conductive layers. The major process trend is compounding and kneading rather than polymerization, with performance governed by rubber pairing, phase-separated morphology, and viscoelastic matching rather than by in situ NBR manufacture.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11347156B2
- Patent Title: Electro-conductive member for electrophotography, process cartridge for ...
- Assignee: Canon Kabushiki Kaisha
- Jurisdiction: US

- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR synthesis or polymerization. It describes manufacture of an electro-conductive member by kneading NBR-based rubber compositions, forming a matrix-domain electro-conductive layer, and molding the composition onto a conductive support followed by evaluation in electrophotographic use. - NBR (1) nitrile content: 15% — Example section, starting materials - NBR (2) nitrile content: 41.5% — Example section, starting materials - NBR (3) nitrile content: 27.5% — Example section, starting materials - NBR (4) nitrile content: 18.0% — Example section, starting materials - NBR (5) nitrile content: 35% — Example section, starting materials - NBR (6) nitrile content: 40.0% — Example section, starting materials - Difference of absolute values of SP values: 0.2 to 4.0 (J/cm³)^{0.5} — Claim 1 - $\tan \delta_1 / \tan \delta_2$: 0.45 to 2.00 — Claim 1 - Electro-conductive particles in domains: 50 to 150 parts by mass per 100 parts by mass of second rubber — Claim 1 - Matrix volume resistivity: $1.0 \times 10^8 \Omega \cdot \text{cm}$ or higher — Claim 1 - Domain volume resistivity: 1.0×10^1 to $1.0 \times 10^4 \Omega \cdot \text{cm}$ — Claim 1 - Temperature: 23° C. — Claim 1 / Example evaluation conditions - Relative humidity: 50% — Claim 1 / Example evaluation conditions - Frequency: 80 Hz — Claim 1 / Example evaluation conditions - Pressure kneader: used for mixing — Example 1 - Carbon black (1): 60 parts by mass — Example 1 - Zinc oxide: 5 parts by mass — Example 1 - Zinc stearate: 2 parts by mass — Example 1 - Polyisoprene: 70 parts by mass — Example 1 - Calcium carbonate: 40 parts by mass — Example 1 - Dispersive sulfur: 3 parts by mass — Example 1 - Di-2-benzothiazolyl disulfide: 2 parts by mass — Example 1 - Tetramethylthiuram disulfide: 0.5 parts by mass — Example 1 - Round bar total length: 252 mm — Example 1 - Round bar outer diameter: 6 mm — Example 1 - Adhesive-coated length: 230 mm — Example 1 - Uncoated end length: 11 mm at both ends — Example 1 - Die inner diameter: 12.5 mm — Example 1 - Electro-conductive support diameter: 5 mm — Example 40 - Outer diameter after polishing: 10.0 mm — Example 40 - Measurement environment: 23° C. and 50% RH for 48 hours — Example 40 - Abutting pressure: 1250 gf (12.26 N) — Example 40 - Output speed: 370 mm/sec — Example 40 - Image resolution: 1,200 dpi — Example 40 - Printing rate: 1.0% — Example 40 - Continuous output: 20000 sheets — Example 40 - Stopped time before evaluation: 12 hours — Example 40 - Halftone images: 20 sheets — Example 40 - Medium & Water Role: No aqueous polymerization medium is disclosed. Water appears only in evaluation conditions as relative humidity (23° C., 50% RH), not as a reaction or formulation medium. - acrylonitrile content: 15%, 18.0%, 27.5%, 35%, 40.0%, and 41.5% nitrile content are disclosed for commercial NBR grades; the low-nitrile NBR (1) is 15%

Medium & Water Role - No aqueous polymerization medium is disclosed. Water appears only in evaluation conditions as relative humidity (23° C., 50% RH), not as a reaction or formulation medium. - Core reaction medium: Solid-state compounding / kneading of rubber compositions; no polymerization medium disclosed

acrylonitrile content - 15%, 18.0%, 27.5%, 35%, 40.0%, and 41.5% nitrile content are disclosed for commercial NBR grades; the low-nitrile NBR (1) is 15%

Relevant Experimental Evidence - Comparative Example 1: EPDM domain and hydrin matrix gave a matrix-domain structure but poor resistance uniformity and rank D image performance. - Comparative Example 3: NBR matrix with EPDM domain formed a matrix-domain structure, but large SP mismatch and poor recovery led to rank D performance. - Comparative Example 5: NBR matrix with SBR domain showed a matrix-domain structure, but accelerator mismatch caused excessive tan δ bias and rank D performance. - Comparative Example 6: NBR used for both domain and matrix failed to form a matrix-domain structure and gave rank D performance. - Example 1: NBR (15% nitrile) was compounded with carbon black, zinc oxide, zinc stearate, sulfur, and accelerators to make an electro-conductive member precursor.

Technical Relevance This patent is relevant because it explicitly discloses NBR grades with low nitrile content, including a 15% nitrile NBR, and shows how NBR behaves in a performance-critical rubber composite. Although it is not a synthesis patent, it provides composition benchmarks and confirms that low-nitrile NBR can be used as a functional rubber phase in engineered formulations.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	acrylonitrile content	Key Findings
US11347156B2	Canon Kabushiki Kaisha	2022	No aqueous polymerization medium is disclosed. Water appears only in evaluation conditions as relative humidity (23° C., 50% RH), not as a reaction or formulation medium.	15%, 18.0%, 27.5%, 35%, 40.0%, and 41.5% nitrile content are disclosed for commercial NBR grades; the low-nitrile NBR (1) is 15%	Comparative Example 1: EPDM domain and hydrin matrix gave a matrix-domain structure but poor resistance uniformity and r

SUPPORTING / RELATED PATENTS (SECONDARY)

Supporting Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about a styrene-acrylonitrile-isoprene triblock copolymer, which is a different block polymer system and not NBR itself.

Supporting Patent 2

- Patent Number: US12247114B1
- Patent Title: Methods of preparing a composite having elastomer and filler
- Assignee: Beyond Lotus Llc
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This is a composite preparation patent listing various elastomers, but it is not about NBR synthesis or NBR as the central invention.

Supporting Patent 3

- Patent Number: US12286702B2
- Patent Title: Method for the production of functionalized elastomeric manufactured articles ...
- Assignee: Wise S.R.L.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about functionalized elastomeric manufactured articles where NBR is only one listed possible support material, not the central invention.

Supporting Patent 4

- Patent Number: US12371558B2
- Patent Title: Resin-extended rubber and process for preparing
- Assignee: Bridgestone Americas Tire Operations, Llc
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about resin-extended rubber compositions using conjugated diene rubbers in non-polar solvent, but it does not specifically disclose NBR as the central invention.

Supporting Patent 5

- Patent Number: EP3902112B1
- Patent Title: Adhesively laminated core for stator, method of manufacturing the same, and ...
- Assignee: Nippon Steel Corporation
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This patent is about an acrylic-based adhesive for laminated stator cores, with NBR only mentioned as a minor elastomer additive.

Supporting Patent 6

- Patent Number: US20230374277A1
- Patent Title: Composite electrolytes with binders
- Assignee: Blue Current, Inc.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about electrolyte composites and binders where NBR appears only as one optional binder among many, so it is not primarily about NBR.

Supporting Patent 7

- Patent Number: US11136459B2
- Patent Title: Viscoelastic conductive nanomaterial-polymer nanocomposites and sensing devices ...
- Assignee: The Provost, Fellows, Foundation Scholars, & the other members of Board, of the ...
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: NBR is only listed as one possible elastomer in a nanocomposite/sensing-device context, so the invention is not primarily about NBR.

Supporting Patent 8

- Patent Number: US10400053B2
- Patent Title: Carbon particles coated with polymer films, methods for their production and ...
- Assignee: Imerys Graphite & Carbon Switzerland Sa
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent concerns carbon particles coated with polymer films, where NBR is only one possible coating polymer in a downstream materials process.

Supporting Patent 9

- Patent Number: EP2561011B1
- Patent Title: Thermoplastic and/or elastomeric composite material based on carbon nanotubes ...
- Assignee: Arkema France
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about a thermoplastic/elastomeric composite material containing acrylonitrile-butadiene-styrene copolymers, not NBR as the central invention.

Supporting Patent 10

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Mitsuboshi Belting Ltd.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about manufacturing a transmission belt core wire and uses carboxy-modified latexes, including an NBR latex derivative, which is a related downstream formulation rather than primary NBR synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The evidence base for this run does not include a true NBR synthesis patent; instead, it provides a formulation-oriented disclosure where NBR is used as a functional rubber phase in an electro-conductive electrophotographic member. The most useful low-acrylonitrile signal is the explicit disclosure of an NBR grade with 15% nitrile content, alongside other grades at 18.0%, 27.5%, 35%, 40.0%, and 41.5%. For low-acrylonitrile NBR work, the patent supports using commercially prepared NBR grades in the ~15–18% nitrile range when balancing elasticity, phase behavior, and electrical performance. No polymerization route, monomer feed ratio, or initiator system is disclosed in the extracted evidence.

5. REFERENCES

- US11347156B2 | Electro-conductive member for electrophotography, process cartridge for ... | Canon Kabushiki Kaisha | US | 2022 | <https://patents.google.com/patent/US11347156B2/en>